

Ashwin Sateesh Kumar

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EXPERIENCE

AI Software Engineer

Dec. 2024 – Present

Citibank

Dallas, TX

- Reduced false positives by 30% and improved detection accuracy by 25% of a ReAct-based multi-agent vulnerability detection system by developing a dependency-aware deep code analyzer with source-to-sink reasoning
- Cut manual penetration testing time by over 50% by building a report generation agent into the ReAct pipeline that delivers structured vulnerability analysis, exploit simulation, and remediation guidance
- Scaled a LangGraph multi-agent security platform to 8,000+ analysts by exposing threat modeling, architecture analysis, and cloud log analysis workflows through a custom MCP server
- Boosted retrieval precision by 25% and cut redundant vector lookups by 40% by architecting a microservices-based RAG system with BM25 + cross-encoder reranking and RAGAS evaluation

Machine Learning Engineer

Jan. 2024 – Dec. 2024

Abecedarian

Boston, MA

- Led the design of a multimodal AI Assistant with novel parallel queue-based continual learning and web scraping for fine-tuning GPT-3.5, enabling real-time updates validated via perplexity score
- Achieved 2× faster image inference while preserving CLIP alignment by distilling a fine-tuned Stable Diffusion model into a lightweight Latent Diffusion Model via U-Net pruning

Research Assistant

July 2023 – Nov. 2023

Northeastern University

Boston, MA

- Uncovered demographic bias (visual and linguistic) in 400k social media posts by training a disentangled multimodal β -VAE with a Mixture of Experts (MoE) fusion layer over U-Net and BERT encoders
- Reduced inference latency by 30% on GCP using bfloat16 quantization, enabling fine-grained control over multimodal generation by manipulating disentangled latent factors to produce debiased representations

Machine Learning Engineer Intern, R&D

June 2022 – Dec. 2022

Signify (Phillips Lighting)

Boston, MA

- Developed a user re-identification system for personalized home automation using an omni-scale feature learning architecture, achieving 0.95 mean average precision (mAP) in identity matching across multi-camera views
- Achieved 95% accuracy in plant anomaly detection, by developing a multispectral vision pipeline with Faster R-CNN localization and U-Net segmentation, driving real-time lighting optimization for crop health monitoring

Trainee Software Engineer

July 2019 – Nov. 2020

KPIT Technologies Ltd.

Bengaluru, India

- Developed an autonomous driving perception system using U-Net traffic scene segmentation and vision-radar sensor fusion pipeline, achieving 0.89 Intersection over Union (IoU) for real-time object detection

TECHNICAL SKILLS

Languages: Python, SQL (PostgreSQL), R

Frameworks: PyTorch, TensorFlow, Scikit-Learn, LangChain, LangGraph, Hugging Face, FastAPI, Flask

Developer Tools: Git, Bash, Docker, GCP (Vertex AI, Cloud Storage), AWS (S3, SageMaker), OpenShift, Tekton

Libraries: NumPy, pandas, SciPy, OpenCV, NLTK, XGBoost, FAISS

PROJECTS

HealthBot | *Python, PyTorch, Flask, HTML, Docker*

Sep. 2023 – Dec. 2023

- Developed a chatbot generating medically relevant responses (0.75 semantic similarity) by fine-tuning FLAN-T5 with LoRA and RLHF (PPO), integrating BERT named entity recognition and disease classification (0.96 F1)

EDUCATION

Northeastern University

Boston, MA

Master of Science in Data Science (GPA: 3.8/4.0)

Sep. 2021 – Dec. 2023

PES University

Bengaluru, India

Bachelor of Engineering in Electronics and Communications

Aug. 2015 – May 2019